

Tutorial Sheet 2

This tutorial sheet is meant to help you remember concepts first covered in Introduction to Aerospace and consolidate your understanding of Flight Dynamics by asking you to carry out a full stability analysis for an existing aircraft. Flight mechanics is not so much about remembering endless formulae (I never can) but instead being able to use your understanding of key concepts to derive any needed expression from first principles. Therefore try this tutorial sheet without looking at the notes.

The Piaggio P.180 Avanti, seen in figure 1, is a twin turboprop executive transport that first flew in the 1980s and features a rather unique three lifting surface design. The lift generated by the fuselage and the inclusion of a lifting canard lead to a more highly loaded wing and a more efficient aerodynamic design. They also make the stability analysis a little bit harder! Key design characteristics for the aircraft can be found in tables 1 and 2. The aircraft is controlled in pitch using an elevator mounted on the horizontal tail. The canard features a set of plain, non-extending flaps, used to maintain trim when the main wing's single slotted, extending flaps are extended and which are not used to control the aircraft.

Static Stability

1. Estimate the whole aircraft lift curve slope and zero-lift angle of attack and drag polar. You may assume elevator deflections to have a minimal effect on lift and drag.
2. From first principles, derive an expression for estimating the aircraft's stick-fixed static margin. You may ignore the effect of the propulsive system on the aircraft's static stability. What is the aftmost allowable position for the CG? Calculate the value of x_{CG} such that the aircraft flies with a static margin of 7.5%.
3. For the CG position identified above and the operating conditions in 2, calculate the elevator deflection necessary to trim the aircraft. Do not forget the effects of thrust.
4. Calculate the canard flap deflection (δ_{fC}) required to keep the aircraft in trim following a deflection of the wing flaps to the landing configuration, $\delta_{fW} = 30^\circ$.
5. The foremost CG position is typically determined based on trimability considerations during landing. Assuming both the canard and wing flaps are deflected as found in Q4, what is the foremost allowable CG position for the Avanti? The maximum landing weight for this aircraft is 4,965 kg and $C'_{L_{max}} = 2.3$.

Dynamic Stability

6. Using the aerodynamic data provided and the CG position from Q2, estimate the U , W , q and $\dot{W}$ derivatives for the Avanti in the equilibrium condition specified in table 2. Try to derive the formulae for each of the 12 derivatives from first principles.

7. Use MATLAB to characterise the dynamic response of the aircraft. Compare the results obtained from analysing the full system to those obtained using the approximate methods in chapter 10 of your notes. Make sure you are able to derive these yourself.
8. To analyse the stick-free dynamic response of the aircraft to a perturbation, the elevator deflection δ_E , must be added as an additional degree of freedom to our system of longitudinal equations of motion. Starting from the longitudinal equations of motion

$$\begin{aligned} m\dot{U} &= X^a - mg \sin \Theta \\ m(\dot{W} - Uq) &= Z^a + mg \cos \Theta \\ I_{yy}\dot{q} &= M^a \end{aligned}$$

and the elevator hinge moment equation

$$I_E \ddot{\delta}_E = H_E = \frac{1}{2} \rho V_\infty^2 S_E c_E (b_0 + b_H \alpha_H + b_E \delta_E + b_T \delta_T),$$

use small-perturbation theory to show that the state-space representation of the aircraft's longitudinal motion starting from an equilibrium, level flight condition, is

$$\begin{bmatrix} m & -X_{\dot{W}} & 0 & 0 & 0 & 0 \\ 0 & m - Z_{\dot{W}} & 0 & 0 & 0 & 0 \\ 0 & -M_{\dot{W}} & I_{yy} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -H_{\dot{W}} & 0 & 0 & I_E & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\Theta} \\ \ddot{\delta}_E \\ \dot{\delta}_E \end{bmatrix} = \begin{bmatrix} X_U & X_W & X_q & -mg & 0 & X_{\delta_E} \\ Z_U & Z_W & Z_q + mU_e & 0 & 0 & Z_{\delta_E} \\ M_U & M_W & M_q & 0 & 0 & M_{\delta_E} \\ 0 & 0 & 1 & 0 & 0 & 0 \\ H_U & H_W & H_q & 0 & 0 & H_{\delta_E} \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \Theta \\ \dot{\delta}_E \\ \delta_E \end{bmatrix}.$$

You may assume that the effect of the elevator's rate of rotation and rotational acceleration on the aircraft's aerodynamics is negligible.

9. Derive expressions for the hinge moment derivatives (H_U , H_W , H_q , $H_{\dot{W}}$, H_{δ_E}) and the elevator control derivatives (X_{δ_E} , Z_{δ_E} , M_{δ_E}). Use the data provided to evaluate these derivatives at the equilibrium condition specified in table 2.
10. Use MATLAB to analyse the dynamic response of the aircraft to perturbations in its states.
 - (a) How does the stick-free response differ from the stick fixed one?
 - (b) How would the addition of a torsion spring, with a torsional stiffness of 10^4 Nm/rad, to the elevator assembly affect the aircraft's response?

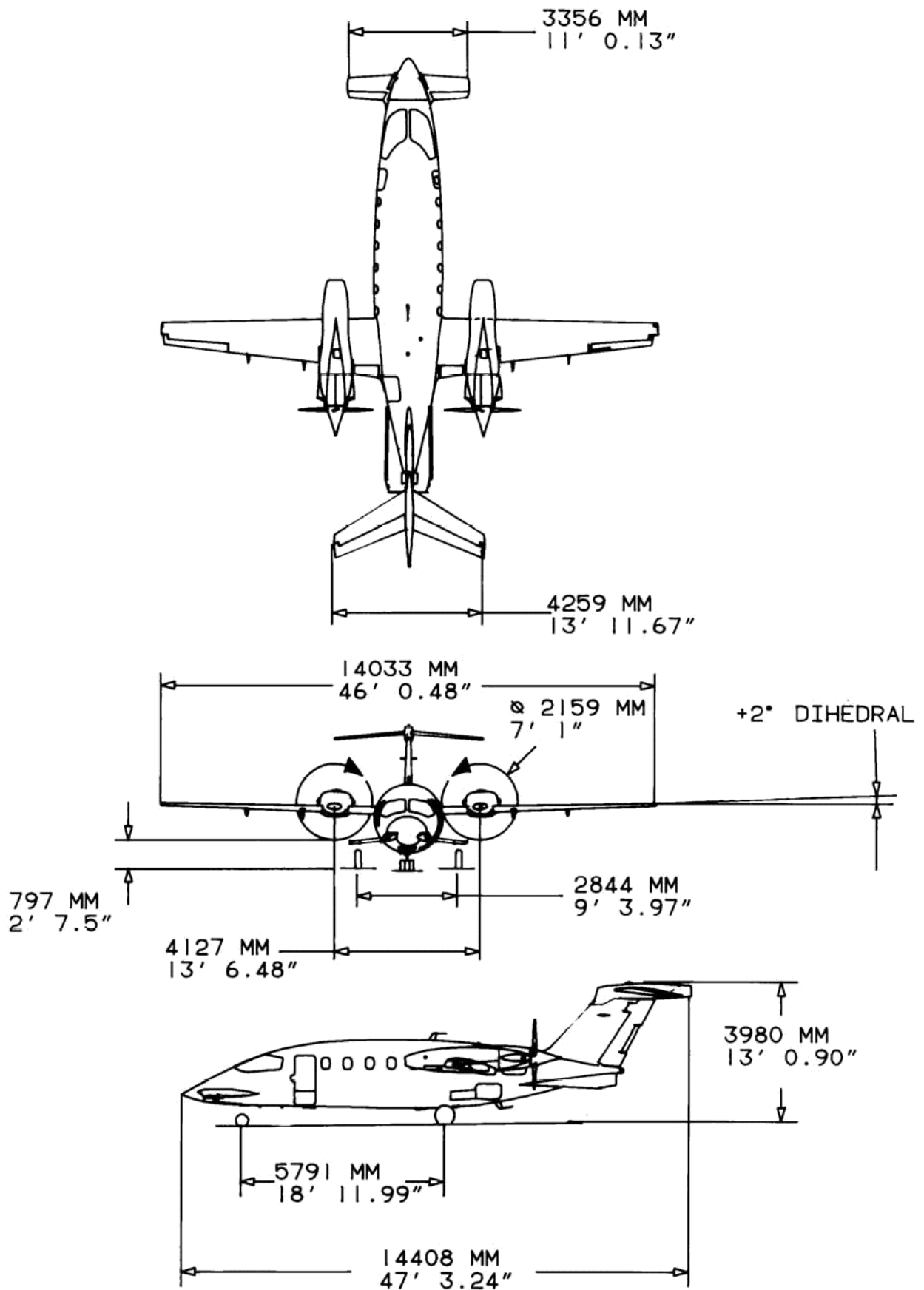


Figure 1: 3-view drawing of the Piaggio P.180 Avanti

Table 1: Lifting Surface Geometric & Aerodynamic Specifications

	Wing	Canard	Horizontal Tail	
Planform area	$S_{ref} = 16.00$	$S_C = 2.19$	$S_H = 3.83$	m ²
Mean Aero Chord	$\bar{c} = 1.18$	$\bar{c}_C = 0.65$	$\bar{c}_H = 0.92$	m
Longitudinal AC position	$x_W = 7.59$	$x_C = 0.62$	$x_H = 13.00$	m
Vertical AC position	$z_W = 0.87$	$z_C = 0.07$	$z_H = 2.99$	m
Aspect ratio	$\mathcal{R}_W = 12.31$	$\mathcal{R}_C = 5.14$	$\mathcal{R}_H = 4.74$	
Quarter Chord Sweep	$\Lambda_W = -0.50$	$\Lambda_C = 0.00$	$\Lambda_H = 30.00$	deg
Lift-curve slope	$a_W = 8.17$	$a_C = 3.40$	$a_H = 4.23$	1/rad
Zero-lift AoA	$\alpha_{0W} = -2.40$	$\alpha_{0C} = -1.40$	$\alpha_{0H} = 2.50$	deg
Setting angle	$i_W = 1.60$	$i_C = 0.00$	$i_H = 0.00$	deg
Zero-lift C_M	$C_{M_{0W}} = -0.46$	$C_{M_{0C}} = -0.0429$	$C_{M_{0H}} = 0.04$	
Up/Downwash gradient	-	$d\varepsilon_U/d\alpha = 0.15$	$d\varepsilon/d\alpha = 0.32$	
Induced Drag k factor	$k_W = 1.1$	$k_C = 2.21$	$k_H = 2.5$	

Table 2: Additional Aerodynamic Characteristics

Fuselage Effects			
Fuselage Pitching Moment Coef	$(dC_M/d\alpha)_F = 0.76$	1/rad	
Drag			
Aircraft Zero-lift Drag Coef	$C_{D_0} = 0.022$		
Wave drag divergence Mach No	$M_{DD} = 0.65$		
Canard Zero Lift Drag Coef	$C_{D_0C} = 0.00685$		
Tailplane Zero-lift Drag Coef	$C_{D_0H} = 0.00951$		
Wing Flaps (Approach Configuration - $\delta_{fW} = 30^\circ$)			
Flapped area ratio	$(S_f/S)_W = 0.55$		
Flap extension ratio	$(c'/c)_W = 1.25$		
Flap zero-lift increment	$\Delta C_{L_0W} = 0.932$		
Flap zero-lift drag increment	$\Delta C_{D_0} = 0.017$		
Canard Flaps			
Flap lift increment	$\Delta C_{L_C}/\delta_{fC} = 1.463$	1/rad	
Elevator			
Elevator to tailplane chord ratio	$c_E/c_H = 0.4$		
Max Elevator deflections	$(\delta_E)_{max} = \pm 20^\circ$		
Elevator lift curve slope	$a_E = 3.08$	1/rad	
Zero-lift Hinge Moment Coef.	$b_0 = 0$	1/rad	
Tailplane Hinge Moment slope	$b_H = -0.502$	1/rad	
Elevator Hinge Moment slope	$b_E = -0.879$	1/rad	
Trim-Tab Hing Moment slope	$b_T = 0$	1/rad	
Propulsion			
Thrust-line position (above baseline)	$z_T = 0.97$	m	
Inertial Propetries at mid-cruise			
Aircraft Mass	$m = 4,836$	kg	
Vertical CoG (above baseline)	$z_{CG} = 0.85$	m	
Pitching 2 nd moment of Inertia	$I_{yy} = 7.529 \times 10^4$	kg m ²	
Elevator polar moment of Inertia	$I_E = 0.261$	kg m ²	
Operating Conditions			
Altitude	$h = 30,000$	ft	
Mach	$M_\infty = 0.62$		

Static Stability

1. Estimate the whole aircraft lift curve slope and zero-lift angle of attack and drag polar.
You may assume elevator deflections to have a minimal effect on lift and drag.

$$C_L = a(\alpha - \alpha_0)$$

Assume elevator deflection is negligible

$$\begin{aligned} C_L &= a_w(\alpha + i_w - \alpha_{0w}) + \frac{S_c}{S_{ref}} a_c \left[\left(1 + \frac{d\epsilon_u}{d\alpha}\right) \alpha + i_c - \alpha_{0c} \right] \\ &\quad + \frac{S_H}{S_{ref}} a_H \left[\left(1 - \frac{d\epsilon}{d\alpha}\right) \alpha + i_H - \alpha_{0H} \right] \\ &= 8.17(\alpha + 4^\circ) + 0.465375(1.15\alpha + 1.4^\circ) + 1.01255625(0.68\alpha - 2.5^\circ) \\ &= 9.4549195\alpha + 0.5375637549 \end{aligned}$$

$$a = \frac{dC_L}{d\alpha} = 9.4549195 \text{ rad}^{-1}$$

$$\text{When } C_L = 0, \alpha_0 = -0.0568554555 \text{ rad} = -3.258^\circ$$

$$\begin{aligned} C_D &= C_{D0} + C_{Di} = C_{D0} + \frac{1}{\pi R_e} C_L^2 \\ &= C_{D0w} + \frac{k_w}{\pi R_w} [a_w(\alpha - \alpha_{0w})]^2 + \frac{S_c}{S_{ref}} \left(C_{D0c} + \frac{k_c}{\pi R_c} \left[a_c \left(1 + \frac{d\epsilon_u}{d\alpha}\right) + i_c - \alpha_{0c} \right]^2 \right) \\ &\quad + \frac{S_H}{S_{ref}} \left(C_{D0H} + \frac{k_H}{\pi R_H} \left[a_H \left(1 - \frac{d\epsilon}{d\alpha}\right) \alpha + i_H - \alpha_{0H} \right]^2 \right) \\ &= C_{D0} + \frac{k_w}{\pi R_w} [a_w(\alpha - \alpha_{0w})]^2 + \frac{S_c}{S_{ref}} \frac{k_c}{\pi R_c} \left[a_c \left(1 + \frac{d\epsilon_u}{d\alpha}\right) + i_c - \alpha_{0c} \right]^2 \\ &\quad + \frac{S_H}{S_{ref}} \frac{k_H}{\pi R_H} \left[a_H \left(1 - \frac{d\epsilon}{d\alpha}\right) \alpha + i_H - \alpha_{0H} \right]^2 \end{aligned}$$

2. From first principles, derive an expression for estimating the aircraft's stick-fixed static margin. You may ignore the effect of the propulsive system on the aircraft's static stability. What is the aftmost allowable position for the CG? Calculate the value of x_{CG} such that the aircraft flies with a static margin of 7.5%.

Ignore the propulsive system. then

$$C_M = \left(\frac{dC_M}{d\alpha} \right)_F \alpha$$

$$+ C_{M0W} - \frac{(x_W - x_{CG})}{\bar{c}} C_{LW} + \frac{(z_W - z_{CG})}{\bar{c}} C_{0W}$$

$$+ \frac{S_c}{S_{ref}} \left[C_{M0c} - (x_c - x_{CG}) C_{Lc} + (z_c - z_{CG}) C_{Dc} \right] \frac{\bar{c}_H}{\bar{c}}$$

$$+ \frac{S_H}{S_{ref}} \left[C_{M0H} - (x_H - x_{CG}) C_{LH} + (z_H - z_{CG}) C_{DH} \right] \frac{\bar{c}_H}{\bar{c}}$$

$$\therefore \frac{dC_M}{dC_L} = - \frac{(x_W - x_{CG})}{\bar{c}} \frac{dC_{LW}}{d\alpha} \frac{d\alpha}{dC_L} + \frac{(z_W - z_{CG})}{\bar{c}} \frac{dC_{0W}}{dC_{LW}} \frac{dC_{LW}}{dC_L} + \dots$$

$$\therefore \frac{dC_{LW}}{dC_L} = \frac{dC_{LW}}{d\alpha} \frac{d\alpha}{dC_L} = \frac{a_W}{a} = 0.8641$$

$$\frac{dC_{0W}}{dC_L} = \frac{dC_{0W}}{dC_{LW}} \frac{dC_{LW}}{dC_L} = 2 \frac{k_W}{\pi R_W} C_{LW} \frac{a_W}{a} = \frac{2k_W a_W C_{LW}}{\pi R_W a}$$

$$\frac{dC_{Lc}}{dC_L} = \frac{dC_{Lc}}{d\alpha} \frac{d\alpha}{dC_L} = \frac{a_c}{a} \left(1 + \frac{d\epsilon_v}{d\alpha} \right) = 0.413541$$

$$\frac{dC_{Dc}}{dC_L} = \frac{dC_{Dc}}{dC_{Lc}} \frac{dC_{Lc}}{dC_L} = 2 \frac{k_c}{\pi R_c} C_{Lc} \frac{a_c}{a} \left(1 + \frac{d\epsilon_v}{d\alpha} \right) = \frac{2k_c a_c C_{Lc}}{\pi R_c a} \left(1 + \frac{d\epsilon_v}{d\alpha} \right)$$

$$\frac{dC_{LH}}{dC_L} = \frac{dC_{LH}}{d\alpha} \frac{d\alpha}{dC_L} = \frac{a_H}{a} \left(1 - \frac{d\epsilon}{d\alpha} \right) = 0.304226$$

$$\frac{dC_{DH}}{dC_L} = \frac{dC_{DH}}{dC_{LH}} \frac{dC_{LH}}{dC_L} = 2 \frac{k_H}{\pi R_H} C_{LH} \frac{a_H}{a} \left(1 - \frac{d\epsilon}{d\alpha} \right) = \frac{2k_H a_H C_{LH}}{\pi R_H a} \left(1 - \frac{d\epsilon}{d\alpha} \right)$$

Since steady-level flight, $T = D$

$$\begin{aligned}
 C_m &= \left(\frac{dC_m}{d\alpha} \right)_F \alpha + (z_T - z_{CG}) C_T + C_{mow} - \frac{x_w - x_{CG}}{\bar{c}} C_{LW} \\
 &\quad + \frac{S_c}{S_{ref}} \left[\frac{\bar{c}_c}{\bar{c}} C_{m_{oc}} - \frac{x_c - x_{CG}}{\bar{c}} C_{Lc} \right] + \frac{S_H}{S_{ref}} \left[\frac{\bar{c}_H}{\bar{c}} C_{m_{oH}} - \frac{x_H - x_{CG}}{\bar{c}} C_{LH} \right] \\
 &= \left(\frac{dC_m}{d\alpha} \right)_F \alpha + (z_T - z_{CG}) C_T + C_{mow} - \frac{x_w - x_{CG}}{\bar{c}} C_{LW} \\
 &\quad - \bar{V}_c C_{Lc} - \bar{V}_H C_{LH}
 \end{aligned}$$

where $\bar{V}_c = \frac{S_c}{S_{ref}} \frac{x_c - x_w}{\bar{c}} = -0.8085$

$\bar{V}_H = \frac{S_H}{S_{ref}} \frac{x_H - x_w}{\bar{c}} = 1.0975$

$$\begin{aligned}
 \therefore \frac{dC_m}{dC_L} &= \frac{0.8641}{1.18} (x_{CG} - 7.59m) + \frac{-0.8085}{0.136875} \cdot 0.413541 (x_{CG} - 0.62m) \\
 &\quad + \frac{1.0975}{0.236275} \cdot 0.304226 (x_{CG} - 13m) \\
 &= \frac{0.73229}{0.73229} x_{CG} - \frac{7.510226}{5.58527}
 \end{aligned}$$

$\therefore$ optimum allowable position, $x_{CG} = x_{np} = \frac{7.510226}{5.58527} m$

for $K_n = -\frac{dC_m}{dC_L} = 7.5\%$

$x_{CG} = \frac{7.510226}{5.58527} m$

3. For the CG position identified above and the operating conditions in 2, calculate the elevator deflection necessary to trim the aircraft. Do not forget the effects of thrust.

To trim the aircraft. $C_m = 0$

$$L = \frac{1}{2} \rho V_{\infty}^2 S_{ref} C_L = W = mg$$

$$\therefore C_L = \frac{mg}{\frac{1}{2} \rho V_{\infty}^2 S_{ref}} = 0.35698$$

Assume elevator deflection is negligible for $C_L = a(\alpha - \alpha_0)$

$$\therefore \alpha = -0.01909944922 \text{ rad} = -1.09432^\circ$$

$$\therefore C_m = \left(\frac{dC_m}{d\alpha} \right)_F \alpha + (z_T - z_{CG}) C_T + C_{mow} - (x_w - x_{CG}) C_{LW}$$

$$\begin{aligned} & -\bar{y}_c C_{Lc} \\ & -\bar{y}_h C_{Lh} \end{aligned} + \frac{S_c \bar{c}_c}{S_{ref} \bar{c}} [C_{m_{oc}} - (x_c - x_{ch}) C_{Lc}] + \frac{S_h \bar{c}_h}{S_{ref} \bar{c}} [C_{m_{oh}} - (x_h - x_{CG}) C_{Lh}] = 0$$

for C_{Lh} simplification

$$\therefore C_{LW} = \underline{a_w (\alpha - \alpha_{ow})} = 0.18618 = C_L - \frac{S_c}{S_{ref}} C_{Lc} - \frac{S_h}{S_{ref}} C_{Lh}$$

$$C_{Lc} = a_c \left[\left(1 + \frac{d\epsilon_c}{d\alpha} \right) \alpha + i_c - \alpha_{oc} \right] = 8.398825945 \times 10^{-3}$$

$$C_{Lh} \text{ (for } C_L) = a_h \left[\left(1 - \frac{d\epsilon_h}{d\alpha} \right) \alpha + i_h - \alpha_{oh} \right] = -0.239506$$

$\therefore D = T$ for trim flight

$$\begin{aligned} \therefore C_D &= C_{D0} + \frac{k_w}{\pi A R_w} C_{LW}^2 + \frac{S_c}{S_{ref}} \frac{k_c}{\pi A R_c} C_{Lc}^2 + \frac{S_h}{S_{ref}} \frac{k_h}{\pi A R_h} C_{Lh}^2 \\ &= C_T = \underline{0.02529} \quad 0.02916 \end{aligned}$$

$$\therefore \frac{S_H \bar{C}_H}{S_{ref} \bar{C}} [C_{MOH} - (\chi_H - \chi_{CG}) C_{LH}] = \underline{0.46739} \quad \underline{0.49219} \quad - \bar{Y}_H C_{LH}$$

$$\therefore C_{LH} = a_H \left[\left(1 - \frac{d\epsilon}{d\alpha} \right) \alpha + i_H - \alpha_{OH} \right] + a_E \delta_E = \underline{-0.456833} \quad - 0.440469$$

$$\therefore \delta_E = \underline{-0.07056 \text{ rad}} \quad - 0.067844 \text{ rad}$$

$$= \underline{-4.0428^\circ} \quad - 3.887^\circ$$

4. Calculate the canard flap deflection (δ_{fC}) required to keep the aircraft in trim following a deflection of the wing flaps to the landing configuration, $\delta_{fW} = 30^\circ$.

For trim flight, $C_m = 0$, when $x_{cg} = 7.488190 \text{ m}$

$$\Delta C_m = \Delta C_{m/w} + \Delta C_{m/c} = 0$$

$$\begin{aligned} \therefore \Delta C_{m/w} &= -\frac{x_w - x_{cg}}{\bar{c}} \Delta C_{L/w} + \frac{z_w - z_{cg}}{\bar{c}} \Delta C_{D_o} \\ &= -\frac{x_w - x_{cg}}{\bar{c}} \left[\Delta C_{L_{0w}} + \frac{S_f}{S_{ref}} a_w \left(\frac{c'}{\bar{c}} - 1 \right) \alpha \right] \\ &= \underline{-0.0800021} - 0.083796 \text{ for } \alpha = 2^\circ \end{aligned}$$

$$\therefore \Delta C_{m/c} = -\frac{x_c - x_{cg}}{\bar{c}} \frac{S_c}{S_{ref}} \Delta C_{Lc} = \underline{0.0800021} \quad 0.083796$$

$$\therefore \Delta C_{Lc} = \underline{0.10044} \quad 0.10518$$

$$\begin{aligned} \therefore \delta_{f/c} &= \Delta C_{Lc} \left(\frac{\Delta C_{Lc}}{\delta_{f/c}} \right)^{-1} = \underline{0.0686555 \text{ rad}} \quad 0.071894 \text{ rad} \\ &= \underline{3.9337^\circ} \quad 4.119^\circ \end{aligned}$$

* C_L also not change, so should construct two equations...

in answer just set $\alpha_e = 2^\circ$

then $\delta_{f/c} = 7.55^\circ$

5. The foremost CG position is typically determined based on trimability considerations during landing. Assuming both the canard and wing flaps are deflected as found in Q4, what is the foremost allowable CG position for the Avanti? The maximum landing weight for this aircraft is 4,965 kg and $C_{L_{max}} = 2.3$.

For CG at foremost position, $C_{L_{max}} = 2.3$, $m = 4965 \text{ kg}$

$$\begin{aligned} \therefore C_L &= a_w (\alpha + i_w - \alpha_{0w}) + \frac{S_f}{S_{ref}} a_w \left(\frac{c'}{c} - 1 \right) \alpha + \Delta C_{L_{0w}} \\ &+ \frac{S_c}{S_{ref}} \left(\frac{\Delta C_{Lc}}{\delta f_c} \delta f_c + a_c \left[\left(1 + \frac{d\epsilon_v}{d\alpha} \right) \alpha + i_c - \alpha_{0c} \right] \right) \\ &+ \frac{S_H}{S_{ref}} a_H \left[\left(1 - \frac{d\epsilon}{d\alpha} \right) \alpha + i_H - \alpha_{0H} \right] \\ &= 10.5171 \alpha + 1.2558108 = 2.3 \end{aligned}$$

$$\therefore \alpha = 0.099285 \text{ rad} = 5.6886^\circ$$

$$\therefore a = \frac{dC_L}{d\alpha} = 10.5171$$

$$\therefore \frac{dC_{Lw}}{dC_L} = \frac{dC_{Lw}}{d\alpha} \frac{d\alpha}{dC_L} = \frac{1}{a} \left[a_w + \frac{S_f}{S_{ref}} a_w \left(\frac{c'}{c} - 1 \right) \right] = 0.88364$$

$$\frac{dC_{Lc}}{dC_L} = \frac{dC_{Lc}}{d\alpha} \frac{d\alpha}{dC_L} = \frac{a_c}{a} \left(1 + \frac{d\epsilon_v}{d\alpha} \right) = 0.37178$$

$$\frac{dC_{LH}}{dC_L} = \frac{dC_{LH}}{d\alpha} \frac{d\alpha}{dC_L} = \frac{a_H}{a} \left(1 - \frac{d\epsilon}{d\alpha} \right) = 0.27350$$

$$\begin{aligned} \therefore \frac{dC_M}{dC_L} &= - \frac{x_w - x_{cg}}{\bar{c}} \frac{dC_{Lw}}{dC_L} - \frac{x_c - x_{cg}}{\bar{c}} \frac{S_c}{S_{ref}} \frac{dC_{Lc}}{dC_L} \\ &- \frac{x_H - x_{cg}}{\bar{c}} \frac{S_H}{S_{ref}} \frac{dC_{LH}}{dC_L} = 0 \text{ for } x_{np} \end{aligned}$$

$$\therefore 7.5844756m - 0.48813645 x_{cg} = 0$$

$$\therefore x_{np} = 15.54786 m$$

$$C_{LW}, C_{LC}, C_{LH} \text{ (for } \delta\epsilon_{max} = -20^\circ)$$

$$\longrightarrow C_m = 0 \text{ for trim (not stable so } \frac{dC_m}{dC_L} \neq 0)$$

$$\longrightarrow x_{cg}$$

Dynamic Stability

6. Using the aerodynamic data provided and the CG position from Q2, estimate the U , W , q and $\dot{W}$ derivatives for the Avanti in the equilibrium condition specified in table 2. Try to derive the formulae for each of the 12 derivatives from first principles.

from 2, $X_{ca} = 7.488190m$

At equilibrium condition, from 3 (actually from answer),

$$\rho = 0.4523 \text{ kg/m}^3, U = C_s M = 303.17 \text{ m/s} \cdot 0.62 = 187.9654 \text{ m/s}$$

$$(C_L)_e = 0.3662, \left(\frac{\partial C_L}{\partial \alpha}\right)_e = a = 9.4549195 \text{ rad}^{-1}$$

$$(C_D)_e = 0.0318 = (C_T)_e, \left(\frac{\partial C_D}{\partial \alpha}\right)_e = \frac{2k}{\pi AR} (C_L)_e = \frac{0.1356 \text{ rad}^{-1}}{0.0208321 \text{ rad}^{-1}}$$

$$(C_M)_e = 0, K_n = 7.5\% = 0.075 \quad \frac{2k_w}{\pi AR_w} (C_{Lw})_e \left(\frac{d\alpha_w}{d\alpha}\right)_e + \dots$$

for small perturbation, U', V', W'

$$X^a = T - D \cos(\alpha - \alpha_e) + L \sin(\alpha - \alpha_e)$$

$$Z^a = -L \cos(\alpha - \alpha_e) - D \sin(\alpha - \alpha_e)$$

$$M^a = -(z_T - z_{ca}) C_T + \frac{1}{2} \rho U_e^2 S_{ref} \bar{c} (C_{M0} - K_n C_L)$$

$$\frac{\partial Y_{\infty}}{\partial U} = 1, \frac{\partial Y_{\infty}}{\partial W} = \frac{\partial Y_{\infty}}{\partial V} = 0$$

$$\frac{\partial \alpha}{\partial U} = \frac{\partial \alpha}{\partial V} = 0, \frac{\partial \alpha}{\partial W} = \frac{1}{U_e}$$

$$\therefore X_u^a = \frac{\partial X}{\partial U} = \frac{\partial X}{\partial Y_{\infty}} \frac{\partial Y_{\infty}}{\partial U} = \frac{\partial X}{\partial Y_{\infty}}$$

$$= \rho U_e S_{ref} [(C_T)_e - (C_D)_e] + \frac{1}{2} \rho U_e^2 S_{ref} \left[\left(\frac{dC_T}{dY_{\infty}}\right)_e - \left(\frac{dC_D}{dY_{\infty}}\right)_e \right]$$

$$= \frac{1}{2} \rho U_e^2 S_{ref} \left[\left(\frac{dC_T}{dV_\infty} \right)_e - \left(\frac{dC_0}{dV_\infty} \right)_e \right]$$

$$\left(\frac{dC_0}{dV_\infty} \right)_e = 0$$

$$\text{for propeller engine, } P = \frac{T V_\infty}{\eta_p} \quad \therefore T = \frac{P \eta_p}{V_\infty}$$

$$\therefore C_T = \frac{P \eta_p}{\left(\frac{1}{2} \rho V_\infty^2 S_{ref} \right) V_\infty} \quad \therefore \frac{dC_T}{dV_\infty} = \frac{-3 P \eta_p}{\frac{1}{2} \rho V_\infty^3 S_{ref}} = -\frac{3}{V_\infty} (C_T)$$

$$\therefore \left(\frac{dC_T}{dV_\infty} \right)_e = -\frac{3}{U_e} (C_T)_e$$

$$\begin{aligned} \therefore X_u^a &= \frac{1}{2} \rho U_e^2 S_{ref} \left[-\frac{3}{U_e} (C_T)_e \right] = -\frac{3}{2} \rho U_e S_{ref} (C_T)_e \\ &= -65.7455 \text{ N/(m/s)} \end{aligned}$$

$$\therefore Z_u^a = \frac{\partial Z}{\partial V_\infty} = -\rho U_e S_{ref} (C_L)_e - \frac{1}{2} \rho U_e^2 S_{ref} \left(\frac{dC_L}{dV_\infty} \right)_e = \dots$$

$$C_L = \frac{C_{L|M=0}}{\sqrt{1-M_\infty^2}} \quad \text{from Prandtl-Glauert}$$

$$\therefore M_u^a = (Z_T - Z_{C_0}) \left(\frac{dC_T}{dV_\infty} \right)_e + \rho U_e S_{ref} (C_M)_e$$

$$+ \frac{1}{2} \rho U_e^2 S_{ref} \left[\left(\frac{dC_{M_0}}{dV_\infty} \right)_e - \left(\frac{dK_n}{dV_\infty} \right)_e (C_L)_e - K_n \left(\frac{dC_L}{dV_\infty} \right)_e \right]$$

$$= - (Z_T - Z_{C_0}) \left(\frac{dC_T}{dV_\infty} \right)_e + \frac{1}{2} \rho U_e^2 S_{ref} \left[\left(\frac{dC_{M_0}}{dV_\infty} \right)_e - K_n \left(\frac{dC_L}{dV_\infty} \right)_e \right] = \dots$$

also = 0

cause $M < M_{00}$

$$\therefore X_w^a = \frac{\partial X}{\partial W} = \frac{\partial X}{\partial \alpha} \frac{\partial \alpha}{\partial W} = \frac{1}{U_e} \frac{\partial X}{\partial \alpha}$$

$$= \frac{1}{2} \rho U_e S_{ref} \left(\frac{dC_L}{d\alpha} \right) e - \frac{1}{2} \rho U_e S_{ref} \left[\left(\frac{dC_D}{d\alpha} \right) e - (C_L) e \right]$$

$$= -\frac{1}{2} \rho U_e S_{ref} \left[\left(\frac{dC_D}{d\alpha} \right) e - (C_L) e \right] = \underline{158.919} \quad \underline{383.89 \text{ N/(m/s)}}$$

$$Z_w^a = -\frac{1}{2} \rho U_e S_{ref} \left[\left(\frac{dC_L}{d\alpha} \right) e + (C_D) e \right] = -6537.833 \text{ N/(m/s)}$$

$$M_w^a = \frac{1}{2} \rho U_e S_{ref} \bar{c} \left[-K_2 \left(\frac{dC_L}{d\alpha} \right) e \right] = -576.66 \text{ Ns}$$

$$U = -(z_c - z_{cw}) q, \quad W = (x_c - x_{cw}) q \quad \text{for canard}$$

$$U = -(z_c - z_{cw}) q, \quad W = (x_c - x_{cw}) q \quad \text{for horizontal tailplane}$$

$$\therefore X_q = (X_q)_c + (X_q)_H$$

$$= (X_u)_c (U_q)_c + (X_u)_H (U_q)_H + (X_w)_c (W_q)_c + (X_w)_H (W_q)_H$$

$$\therefore Z_q = (Z_u)_c (U_q)_c + (Z_u)_H (U_q)_H + (Z_w)_c (W_q)_c + (Z_w)_H (W_q)_H$$

$$\therefore M_q = (x_H - x_{cw}) Z_{qH} - (z_H - z_{cw}) X_{qH}$$

$$(x_c - x_{cw}) Z_{qc} - (z_c - z_{cw}) X_{qc}$$

$$+ (M_u)_c (U_q)_c + (M_u)_H (U_q)_H + (Z_u)_c (U_q)_c + (Z_u)_H (U_q)_H$$

$$\begin{aligned}\epsilon &= \epsilon_0 + \frac{d\epsilon}{d\alpha} \alpha \left(1 - \frac{l_H}{U_e} \right) = \epsilon_0 + \frac{d\epsilon}{d\alpha} \left(\alpha - \frac{d\alpha}{dt} \frac{l_H}{U_e} \right) \\ &= \epsilon_0 + \frac{d\epsilon}{d\alpha} \left(\alpha - \frac{l_H}{U_e^2} \dot{W} \right)\end{aligned}$$

$$\therefore \frac{d\epsilon}{d\dot{W}} = - \frac{d\epsilon}{d\alpha} \frac{l_H}{U_e^2}$$

$$\therefore X_{\dot{W}} = (X_W)_H (E_W)^{-1} (E_W) = \frac{l_H}{U_e} \frac{d\epsilon}{d\alpha} (X_W)_H$$

$$\therefore Z_{\dot{W}} = \frac{l_H}{U_e} \frac{d\epsilon}{d\alpha} (Z_W)_H$$

$$\therefore M_{\dot{W}} = \frac{l_H}{U_e} \frac{d\epsilon}{d\alpha} (M_W)_H - (Z_H - Z_{CG}) X_{\dot{W}} + (X_H - X_{CG}) Z_W$$

7. Use MATLAB to characterise the dynamic response of the aircraft. Compare the results obtained from analysing the full system to those obtained using the approximate methods in chapter 10 of your notes. Make sure you are able to derive these yourself.

for SPPD, ...

for Phogoid, $\omega_d = \frac{\sqrt{2}g}{U_e}$, $\zeta = \frac{1}{\sqrt{2(L/D)_e}}$

see solution

8. To analyse the stick-free dynamic response of the aircraft to a perturbation, the elevator deflection δ_E , must be added as an additional degree of freedom to our system of longitudinal equations of motion. Starting from the longitudinal equations of motion

$$\begin{aligned} m\dot{U} &= X^a - mg \sin \Theta \\ m(\dot{W} - Uq) &= Z^a + mg \cos \Theta \\ I_{yy}\dot{q} &= M^a \end{aligned}$$

and the elevator hinge moment equation

$$I_E \ddot{\delta}_E = H_E = \frac{1}{2} \rho V_\infty^2 S_E c_E (b_0 + b_H \alpha_H + b_E \delta_E + b_T \delta_T),$$

use small-perturbation theory to show that the state-space representation of the aircraft's longitudinal motion starting from an equilibrium, level flight condition, is

$$\begin{bmatrix} m & -X_{\dot{W}} & 0 & 0 & 0 & 0 \\ 0 & m - Z_{\dot{W}} & 0 & 0 & 0 & 0 \\ 0 & -M_{\dot{W}} & I_{yy} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -H_{\dot{W}} & 0 & 0 & I_E & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{W} \\ \dot{q} \\ \dot{\Theta} \\ \ddot{\delta}_E \\ \dot{\delta}_E \end{bmatrix} = \begin{bmatrix} X_U' & X_W & X_q & -mg & 0 & X_{\delta_E} \\ Z_U & Z_W & Z_q + mU_e & 0 & 0 & Z_{\delta_E} \\ M_U & M_W & M_q & 0 & 0 & M_{\delta_E} \\ 0 & 0 & 1 & 0 & 0 & 0 \\ H_U & H_W & H_q & 0 & 0 & H_{\delta_E} \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} U \\ W \\ q \\ \Theta \\ \dot{\delta}_E \\ \delta_E \end{bmatrix}.$$

You may assume that the effect of the elevator's rate of rotation and rotational acceleration on the aircraft's aerodynamics is negligible.

$$\mathbf{x} = [U \ W \ q \ \Theta \ \delta_E \ \dot{\delta}_E]^T$$

$$\therefore \mathbf{M}\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} = \mathbf{N}(\mathbf{x}) + \mathbf{F}(\mathbf{x}, \dot{\mathbf{x}}) + \mathbf{G}(\mathbf{x})$$

$$\text{At equilibrium, } \dot{\mathbf{x}}_0 = \mathbf{N}(\mathbf{x}_0) + \mathbf{F}(\mathbf{x}_0, \dot{\mathbf{x}}_0) + \mathbf{G}(\mathbf{x}_0) = \mathbf{0}$$

$$\therefore \mathbf{M}(\ddot{\mathbf{x}}_0 + \dot{\mathbf{x}}') = \mathbf{x}' \mathbf{N}_{\mathbf{x}_0} + \mathbf{x}' \mathbf{F}_{\mathbf{x}_0} + \mathbf{x}' \mathbf{F}_{\dot{\mathbf{x}}_0} + \mathbf{x}' \mathbf{G}_{\mathbf{x}_0}$$

$$\therefore (\mathbf{M} - \mathbf{F}_{\dot{\mathbf{x}}_0}) \dot{\mathbf{x}}' = \mathbf{M}_* \dot{\mathbf{x}}' = \mathbf{x}' (\mathbf{N}_{\mathbf{x}_0} + \mathbf{F}_{\mathbf{x}_0} + \mathbf{G}_{\mathbf{x}_0})$$

$$\text{Let } \mathbf{x}' = \mathbf{x}^* e^{\lambda t}$$

$$\therefore |-\lambda(\mathbf{M} - \mathbf{F}_{\dot{\mathbf{x}}_0}) + \mathbf{N}_{\mathbf{x}_0} + \mathbf{F}_{\mathbf{x}_0} + \mathbf{G}_{\mathbf{x}_0}| = 0$$

Apply small perturbation, then

$$m(\ddot{U} + \ddot{U}') = X^a + U'X_u + W'X_w + q'X_q - \Theta'mg \cos\Theta + \delta_E' X_{\delta_E} + \dot{W}'X_{\dot{W}}$$

$$m(\ddot{W} + \ddot{W}') = m(U_0 + U') \overbrace{(q_e + q')}^0$$

$$= Z^a + U'Z_u + W'Z_w + q'Z_q - \Theta'mg \sin\Theta + \delta_E' Z_{\delta_E} + \dot{W}'Z_{\dot{W}}$$

$$I_E(\ddot{\delta}_E + \ddot{\delta}_E') = H_E + U'H_E + W'H_E + q'H_E + \delta_E'H_E + \dot{W}'H_E$$

$$\text{and } \dot{\delta}_E + \dot{\delta}_E' = \dot{\delta}_E + \dot{\delta}_E'$$

see Review 3 for further information

9. Derive expressions for the hinge moment derivatives (H_U , H_W , H_q , $H_{\dot{W}}$, H_{δ_E}) and the elevator control derivatives (X_{δ_E} , Z_{δ_E} , M_{δ_E}). Use the data provided to evaluate these derivatives at the equilibrium condition specified in table 2.

$$H_U = \frac{dH_E}{dU} = \frac{dH_E}{dV_\infty} \frac{dV_\infty}{dU} = \frac{dH_E}{dV_\infty}$$

$$= \rho V_\infty S_E C_E (b_0 + b_H \alpha_H + b_E \delta_E + b_T \dot{\delta}_T) = 0 \text{ initially equilibrium}$$

$$\alpha_H = (1 - \frac{d\epsilon}{d\alpha}) \alpha + i_H - \alpha_{OH}$$

$$H_W = \frac{dH_E}{dW} = \frac{1}{U_e} \frac{dH_E}{d\alpha} = \frac{1}{2} \rho V_\infty S_E C_E b_H (1 - \frac{d\epsilon}{d\alpha})$$

$$U = -(z_H - z_{CG}) q$$

$$W = (x_H - x_{CG}) q$$

$$H_q = H_U U_q + H_W W_q$$

$$H_{\delta_E} = \frac{1}{2} \rho V_\infty^2 S_E C_E b_E$$

$$H_{\dot{W}} = H_W \frac{d\alpha}{d\alpha_H} \frac{d\alpha_H}{d\epsilon} - \frac{l_H}{U_e^2} (\frac{d\epsilon}{d\alpha}) = \frac{l_H}{U_e} (\frac{d\epsilon}{d\alpha}) H_W$$

see Review 3 for further information

10. Use MATLAB to analyse the dynamic response of the aircraft to perturbations in its states.
- (a) How does the stick-free response differ from the stick fixed one?
 - (b) How would the addition of a torsion spring, with a torsional stiffness of 10^4 Nm/rad, to the elevator assembly affect the aircraft's response?

see solutions...